

Frenet-Refine: Differentiable Septuple Diffusion for Tubular Topology Refinement

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Abstract

Accurate segmentation of tubular structures (e.g., vessels, airways, neurons) is plagued by a fundamental trade-off: discriminative models excel at pixel-wise accuracy but suffer from topological breakage under occlusion, while generative models capture global geometry but struggle with local alignment. In this paper, we propose **Frenet-Refine**, a hybrid framework that bridges this gap by treating segmentation as a **coarse-to-fine refinement** process on the Riemannian manifold of rigid motions.

We first leverage a lightweight discriminative network, supervised with differentiable topology and curvature losses, to generate a coarse structural proposal. Crucially, instead of post-processing this output in voxel space, we **lift** the fragmented skeleton onto the **SE(3) fiber bundle**. We then apply a novel **SE(3)-equivariant graph diffusion refiner** which treats the coarse proposal as a noisy signal and “denoises” it using learned anatomical priors. This refiner acts as a geometric hallucinator, seamlessly reconnecting broken branches and smoothing zig-zag artifacts that no pixel-wise loss can resolve.

Experiments on challenging airway and vessel benchmarks show that Frenet-Refine reduces topological errors by more than 40% compared to strong topology-aware baselines and diffusion-based segmentors, while adding only a small inference overhead. Our results suggest a general recipe for SE(3)-guided refinement that can complement modern medical foundation models and improve the structural fidelity of thin-structure segmentation.

1 Introduction

In medical image analysis, the topology of tubular structures—connectivity, branching, and loops—is often more clinically significant than pixel-level overlap. A vessel segmentation with 99% Dice score can be functionally useless for hemodynamic simulation, surgical path planning, or stenosis quantification if it contains a single critical disconnection.

Current approaches to tubular segmentation fall into two distinct paradigms:

1. The Discriminative Paradigm (Bottom-Up). Methods like nnU-Net [?], SwinUNETR [?], and topology-loss-guided networks (e.g., clDice [?], TopoLoss [?], Skeleton Recall Loss [?], and Frenet-TopoNet [?]) classify voxels based on local texture evidence. While precise, they are inherently brittle: when the image signal is weak (e.g., due to severe stenosis, metal artifacts, or low contrast), the model lacks an explicit geometric prior to infer the missing connection, resulting in broken topology.

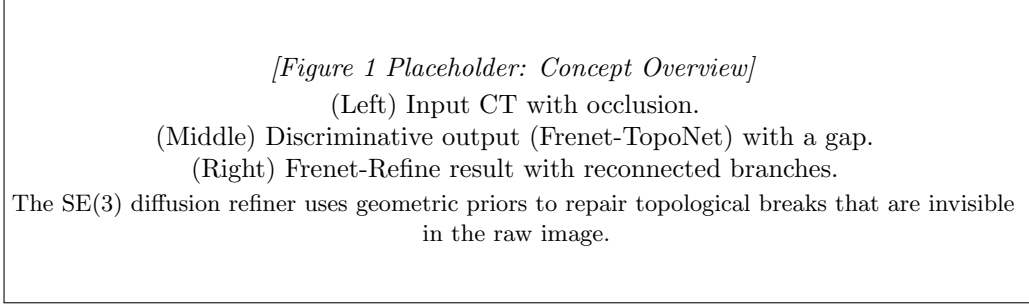


Figure 1: **Core idea of Frenet-Refine.** Instead of voxel-level inpainting, we perform topology repair on the SE(3) manifold of framed centerlines.

2. The Generative Paradigm (Top-Down). Recent geometric generative models, including diffusion-based segmentors [? ? ?] and SE(3)-based curve generators [? ?], can synthesize anatomically plausible vessel graphs from noise. However, constraining these models to align perfectly with a specific patient’s CT scan is computationally expensive and prone to posterior collapse, often ignoring fine pathological details.

Meanwhile, medical foundation models such as MedSAM-2 [?], STU-Net [?], and U-Mamba [?] have pushed Dice performance close to saturation across many organs, but still operate primarily in voxel space and lack explicit mechanisms for repairing hard occlusions in thin structures.

We ask: *Can we combine the pixel-level precision of discriminative models with the structural robustness of geometric generative priors?*

To answer this, we propose **Frenet-Refine**, a framework that introduces a paradigm shift from “voxel inpainting” to “**manifold geometric refinement.**” Our key insight is that topological breaks are not merely pixel errors, but *geometric noise* on the underlying fiber bundle of SE(3)-valued curve frames.

Our contributions are three-fold:

- **Generative-discriminative synergy.** We propose a two-stage framework where a discriminative segmentor provides a coarse proposal, and a conditional SE(3) diffusion model refines it. This solves hard occlusion problems that no discriminative loss alone can address.
- **SE(3) lifting and refinement.** We formulate the refinement task not in the voxel grid, but on the SE(3) manifold. By lifting the skeleton to a graph of rigid frames and using an E(n)-equivariant graph transformer, we enforce geometric validity (smoothness, orthogonality, and SE(3)-equivariance) by design.
- **SOTA topological fidelity.** On challenging airway and vessel datasets, Frenet-Refine outperforms purely discriminative SOTA methods (including Frenet-TopoNet, DeformCL, and Topo-Mamba) as well as pixel-space diffusion baselines in topological metrics (Betti error and Gap Closure Rate), bridging the gap between segmentation and geometric modeling.

2 Related Work

2.1 Topology-Aware Segmentation

Standard segmentation networks optimize pixel-wise losses (Dice / cross-entropy) and often ignore global structure. To address this, topology-aware methods introduce auxiliary losses based on centerlines or persistent homology. `clDice` [?] enforces centerline overlap; `TopoLoss` [?] minimizes Betti number mismatch via persistent homology; `Skeleton Recall Loss` [?], `Centerline Boundary Dice` [?], and curvature-aware models [?] further constrain thin structures. Recently, **Frenet-TopoNet** [?] proposed a differentiable centerline field with Frenet curvature regularization for anatomically valid tubular trees. While these methods improve connectivity in noisy regions, they fundamentally rely on image evidence: if the signal is entirely missing (“hard occlusion”), the loss function has no gradient to pull the disconnected components together.

2.2 Medical Foundation Models and State-Space Segmentors

Large medical foundation models such as `MedSAM-2` [?], `STU-Net` [?], and `U-Mamba` [?] provide strong, often near-saturated Dice performance across many organs. For thin structures, `Topo-Mamba` [?] and deformable-centerline methods [?] encode topology priors or centerline dynamics into state-space or transformer backbones. These approaches still operate within the voxel grid; topology repair is implemented as a local regularizer rather than an explicit geometric editing process. In contrast, `Frenet-Refine` treats segmentation refinement as $SE(3)$ -guided generative editing on a low-dimensional curve manifold, and can be applied on top of any backbone, including foundation models.

2.3 Generative Models for Segmentation and Editing

Diffusion models [?] have recently been applied to medical segmentation as implicit priors. `MedSegDiff` [?], `SegDiff` [?], and related works [?] generate segmentation masks in pixel space, while `SDEdit` and `RePaint` [?] perform guided image editing. However, these methods operate in the high-dimensional voxel space, which is inefficient for sparse tubular structures and does not explicitly maintain the “curve + frame” structure of anatomy.

Our work adapts these ideas to a geometric setting: we build on $SE(3)$ -equivariant diffusion on fiber bundles [?], score-based graph generation [?], and geometric deep learning [?] to perform refinement on a graph of $SE(3)$ nodes.

2.4 Datasets and Applications

We evaluate on `ATM’22` for airway tree modeling [?], `TubeTK` for intracranial vessels [?], and discuss extension to other tubular domains such as dental root canals [?] and anatomical trees [?]. While our experiments focus on medical imagery, the proposed $SE(3)$ -guided refinement recipe is broadly applicable to any 1D manifold embedded in 3D (e.g., robot trajectories, road networks, and neural fibers).

3 Method

Our pipeline, illustrated in Figure 2, consists of two cascaded stages: (1) **proposal generation** via a topology-aware discriminative network, and (2) **structural refinement** via conditional SE(3) graph diffusion.

3.1 Problem Formulation

Let $I \in \mathbb{R}^{H \times W \times D}$ be a 3D image volume, and let $\mathcal{C}^* \subset \mathbb{R}^3$ denote the (unknown) patient-specific tubular manifold (e.g., centerline tree). The ideal segmentation corresponds to a mask S^* and associated centerline graph $G^* = (\mathcal{V}^*, \mathcal{E}^*)$ that faithfully discretize $\mathcal{C}^*$, preserving topology (Betti numbers) and geometric smoothness (bounded curvature/torsion) [? ?].

We view segmentation as recovering G^* given I :

$$G^* = \arg \max_G p(G | I), \quad G \in \mathcal{G}_{\text{tubular}}, \quad (1)$$

where $\mathcal{G}_{\text{tubular}}$ is the family of valid tubular graphs.

Frenet-Refine adopts a coarse-to-fine approximation:

$$G^* \approx \text{Refine}_\phi(G_{\text{coarse}}^{\text{SE}(3)}, I), \quad (2)$$

where $G_{\text{coarse}}^{\text{SE}(3)}$ is an SE(3)-valued skeleton graph produced by a discriminative segmentor and lifted onto the SE(3) fiber bundle, and Refine_ϕ is a learned SE(3) diffusion refiner.

3.2 Stage 1: Coarse Proposal Generation

The goal of the first stage is to extract as much structure as possible from the image evidence. We employ a 3D U-Net backbone or a strong medical foundation segmentor (e.g., STU-Net [?] or U-Mamba [?]), trained with the Frenet-TopoNet objective [?], which we briefly review.

The network predicts a probability map $P = f_\theta(I)$, from which we derive a coarse segmentation S_{coarse} . We supervise P with a joint loss:

$$\mathcal{L}_{\text{Stage1}} = \mathcal{L}_{\text{Dice}} + \lambda_{\text{topo}} \mathcal{L}_{\text{PH}} + \lambda_{\text{curv}} \mathcal{L}_{\text{Frenet}}, \quad (3)$$

where $\mathcal{L}_{\text{PH}}$ minimizes persistent homology errors (Betti numbers) [? ?] and $\mathcal{L}_{\text{Frenet}}$ enforces local smoothness via curvature/torsion penalties [? ?]. This stage outputs a coarse segmentation S_{coarse} that is accurate in high-contrast regions but often contains gaps where the vessel signal is weak or occluded.

3.3 The Bridge: Lifting to the SE(3) Fiber Bundle

To repair topological defects, we move away from the voxel grid and operate on a geometric graph representation, a process we call **lifting**.

1. Skeletonization. We extract a centerline graph $G_{\text{skel}} = (\mathcal{V}, \mathcal{E})$ from S_{coarse} using a differentiable soft-skeletonization operator [? ?]. Each node $v_i \in \mathcal{V}$ has a position $\mathbf{x}_i \in \mathbb{R}^3$.

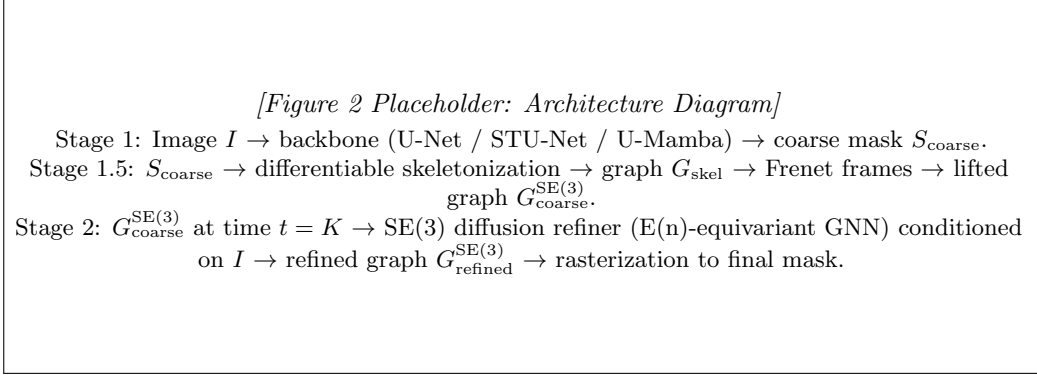


Figure 2: **Pipeline of Frenet-Refine.** A topology-aware discriminative model provides a coarse mask, which is lifted onto the SE(3) fiber bundle and refined via conditional diffusion.

2. Frenet frame estimation. For each node, we estimate a local Frenet frame $\mathbf{R}_i = [\mathbf{N}_i, \mathbf{B}_i, \mathbf{T}_i] \in \text{SO}(3)$:

$$\mathbf{T}_i = \frac{\mathbf{x}_{i+1} - \mathbf{x}_{i-1}}{\|\mathbf{x}_{i+1} - \mathbf{x}_{i-1}\|}, \quad (4)$$

$$(\mathbf{N}_i, \mathbf{B}_i) = \text{PCA-normal-plane}(\{\mathbf{x}_j\}_{j \in \mathcal{N}(i)}), \quad (5)$$

where $\mathcal{N}(i)$ is a small neighborhood along the centerline. This yields an orthonormal triad aligned with the curve geometry [? ?].

3. Fiber bundle construction. The lifted node state is

$$\mathbf{g}_i = (\mathbf{R}_i, \mathbf{x}_i) \in \text{SE}(3), \quad (6)$$

and the lifted graph $G_{\text{coarse}}^{\text{SE}(3)} = (\mathcal{V}, \mathcal{E}, \{\mathbf{g}_i\})$ resides on the SE(3) fiber bundle. Intuitively, each node is a rigid body representing a local cross-section of the tubular structure.

Under random SE(3) augmentations of the input image, both the backbone and the lifting operator are approximately SE(3)-equivariant, and our refiner preserves this equivariance by construction through equivariant message passing [? ?].

3.4 Stage 2: SE(3) Generative Refinement

We treat the coarse graph $G_{\text{coarse}}^{\text{SE}(3)}$ as a *corrupted observation* of the true latent anatomy. Our goal is to “denoise” this graph on the SE(3) manifold to recover the ground-truth connectivity.

Conditional SDE on SE(3). We build on SE(3)-valued diffusion models [? ?] formulated as stochastic differential equations (SDEs) in the Lie algebra $\mathfrak{se}(3)$:

$$d\mathbf{g}_t = \mathbf{g}_t \circ d\boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \in \mathfrak{se}(3), \quad (7)$$

where $\circ$ denotes group composition and $d\boldsymbol{\xi}_t$ is Brownian motion in $\mathfrak{se}(3)$.

Following score-based generative modeling [?], we learn a time-dependent score network s_ϕ that approximates the gradient of the log density:

$$s_\phi(\mathbf{g}_t, t, I) \approx \nabla_{\mathbf{g}_t} \log p_t(\mathbf{g}_t | I), \quad (8)$$

parameterized by an E(n)-equivariant graph transformer [? ?] operating on SE(3) node states with pooling along edges.

Refinement as SDEdit on the manifold. Instead of generating from pure noise at $t = T$, we employ an SDEdit-style *editing* procedure on SE(3):

- **Forward (corruption).** We interpret $G_{\text{coarse}}^{\text{SE}(3)}$ as an approximate sample from the posterior at an intermediate time $t = 0$. We then apply the forward SDE for a short duration to reach $t = K$:

$$G_K^{\text{SE}(3)} = \text{ForwardSDE}(G_{\text{coarse}}^{\text{SE}(3)}, 0 \rightarrow K). \quad (9)$$

This “loosens” the erroneous structure (e.g., false endpoints at a gap) while preserving large-scale geometry.

- **Reverse (refinement).** Starting from $G_K^{\text{SE}(3)}$, we solve the reverse-time SDE from $t = K$ down to $t = 0$:

$$d\mathbf{g}_t = \mathbf{g}_t \circ \left[-\beta(t)s_\phi(\mathbf{g}_t, t, I) dt + \sqrt{\beta(t)} d\bar{\mathbf{w}}_t \right], \quad (10)$$

where $\beta(t)$ is a noise schedule and $d\bar{\mathbf{w}}_t$ is reverse-time noise. This yields a refined graph $G_{\text{refined}}^{\text{SE}(3)}$.

Prior–likelihood decomposition. We explicitly decompose the score into a geometric prior and image-guided likelihood term:

$$s_\phi(\mathbf{g}_t, t, I) = s_{\text{prior}}(\mathbf{g}_t, t) + \gamma s_{\text{likelihood}}(\mathbf{g}_t, t, I), \quad (11)$$

mirroring the Bayesian factorization $p(G | I) \propto p(G)p(I | G)$. Here, s_{prior} enforces geometric smoothness, Frenet consistency, and tubular connectivity; $s_{\text{likelihood}}$ pulls the graph towards weak image evidence (e.g., low-contrast vessels) using cross-attention to multi-scale image features. The scalar γ controls the trade-off between hallucinating missing branches and adhering strictly to image evidence.

3.5 Training Strategy: Two-Stage Pretraining and End-to-End Refinement

We adopt a two-stage training scheme followed by an end-to-end refinement fine-tuning. This preserves the benefits of a strong geometric prior while allowing gradients to flow through the refinement pipeline.

Stage A: Discriminative and generative pretraining. We first train the Stage 1 segmentor and Stage 2 SE(3) refiner separately:

- **Stage 1 pretraining.** The backbone (U-Net / STU-Net / U-Mamba / Frenet-TopoNet) is trained with $\mathcal{L}_{\text{Stage1}}$ on the coarse mask S_{coarse} , as described in Sec. 3.2.
- **Stage 2 pretraining.** The SE(3) refiner is trained as a score-based diffusion model on clean ground-truth graphs $G_{\text{clean}}^{\text{SE}(3)}$ obtained from the annotation masks, using denoising score matching in the Lie algebra $\mathfrak{se}(3)$. Concretely, for each sample we sample $t \sim \mathcal{U}(0, T)$, perturb $G_{\text{clean}}^{\text{SE}(3)}$ to G_t with the forward SDE, and minimize

$$\mathcal{L}_{\text{score}} = \mathbb{E}_{G_{\text{clean}}^{\text{SE}(3)}, t} \left[\lambda(t) \|s_\phi(G_t, t, I) - \hat{\nabla}_{G_t} \log p(G_t | G_{\text{clean}}^{\text{SE}(3)})\|^2 \right], \quad (12)$$

where $\hat{\nabla}$ denotes the standard noise-prediction surrogate used in diffusion models [? ?].

After Stage A, we obtain a strong discriminative segmentor and a geometry-aware SE(3) prior that are trained independently.

Stage B: End-to-end refinement fine-tuning. We then connect the two modules and fine-tune the whole pipeline end-to-end. For each training image I with ground-truth mask S^* :

1. Forward I through the Stage 1 segmentor to obtain S_{coarse} .
2. Skeletonize and lift S_{coarse} to obtain $G_{\text{coarse}}^{\text{SE}(3)}$.
3. Apply a short SE(3) refinement with a small number of reverse steps ($N_{\text{ref}} \approx 5-10$), starting from $G_{\text{coarse}}^{\text{SE}(3)}$ at time $t = K$ and using the pretrained score network s_ϕ . In Stage B we use a deterministic Euler update (ODE-style) for stability:

$$\mathbf{g}^{k-1} = \mathbf{g}^k \circ \exp(-\Delta t \beta_k s_\phi(\mathbf{g}^k, t_k, I)), \quad (13)$$

where $\exp$ is the SE(3) exponential map and $k = 1, \dots, N_{\text{ref}}$.

4. Rasterize the refined graph $G_{\text{refined}}^{\text{SE}(3)}$ back to a refined mask S_{refined} using a differentiable tubular renderer.

We then define a joint loss

$$\mathcal{L}_{\text{E2E}} = \mathcal{L}_{\text{coarse}}(S_{\text{coarse}}, S^*) + \alpha \mathcal{L}_{\text{refined}}(S_{\text{refined}}, S^*), \quad (14)$$

where

$$\mathcal{L}_{\text{coarse}} = \mathcal{L}_{\text{Dice}}(S_{\text{coarse}}, S^*) + \lambda_{\text{topo}} \mathcal{L}_{\text{PH}}(S_{\text{coarse}}, S^*), \quad (15)$$

$$\mathcal{L}_{\text{refined}} = \mathcal{L}_{\text{Dice}}(S_{\text{refined}}, S^*) + \lambda_{\text{topo}} \mathcal{L}_{\text{PH}}(S_{\text{refined}}, S^*) + \lambda_{\text{curv}} \mathcal{L}_{\text{Frenet}}(S_{\text{refined}}, S^*). \quad (16)$$

The scalar α balances how strongly we supervise the refined output versus the coarse prediction.

During Stage B, we backpropagate gradients through the lifting, the SE(3) refinement steps, and back into the Stage 1 segmentor. In practice, we keep s_ϕ either frozen or fine-tune it with a much smaller learning rate, and use a fixed number of reverse ODE steps, which makes the whole refinement process differentiable and computationally tractable.

Algorithm 1 Frenet-Refine Inference-Time Refinement (SE(3) Editing)

- 1: **Input:** Image I , trained Stage 1 segmentor f_θ , trained SE(3) refiner s_ϕ .
 - 2: **Output:** Refined mask $\hat{S}$.
 - 3: $P \leftarrow f_\theta(I)$; obtain coarse mask S_{coarse} by thresholding.
 - 4: Skeletonize S_{coarse} to obtain G_{skel} .
 - 5: Lift G_{skel} to SE(3) graph $G_{\text{coarse}}^{\text{SE}(3)}$.
 - 6: Apply forward SDE from $t = 0$ to $t = K$ to get $G_K^{\text{SE}(3)}$.
 - 7: Use score network s_ϕ to simulate reverse SDE/ODE from $t = K$ to $t = 0$ and obtain $G_{\text{refined}}^{\text{SE}(3)}$.
 - 8: Rasterize $G_{\text{refined}}^{\text{SE}(3)}$ back to a binary mask $\hat{S}$.
-

Inference-time behavior. At test time, we use the same refinement procedure as in Stage B, but without ground-truth supervision. The only difference is that the refinement is run once, using the final parameters of both Stage 1 and Stage 2 learned in the end-to-end phase.

3.6 Geometric View: Curve Evolution on the SE(3) Bundle

So far we have described Frenet-Refine procedurally. We now provide a geometric interpretation that connects our design choices to classical curve evolution and elastic-rod models [? ?].

From discrete centerlines to elastic rods. Let $\gamma(s) \in \mathbb{R}^3$ be a smooth centerline parameterized by arc length s . Its Frenet frame $(\mathbf{T}(s), \mathbf{N}(s), \mathbf{B}(s))$ and curvature/torsion $(\kappa(s), \tau(s))$ uniquely characterize local shape [?]. Classical models of elastic rods [?] define an energy

$$\mathcal{E}_{\text{rod}}(\gamma) = \int (\lambda_\ell + \lambda_\kappa \kappa(s)^2 + \lambda_\tau \tau(s)^2) ds, \quad (17)$$

whose gradient flow shortens the curve while penalizing sharp bends and excessive twisting.

In our discrete setting, the lifted SE(3) graph $G^{\text{SE}(3)} = (\mathcal{V}, \mathcal{E}, \{\mathbf{g}_i\})$ can be viewed as a polygonal approximation of γ , where each node state $\mathbf{g}_i = (\mathbf{R}_i, \mathbf{x}_i)$ encodes a local frame and cross-section. Discrete curvature and torsion can be computed from consecutive SE(3) increments, and our Frenet loss $\mathcal{L}_{\text{Frenet}}$ is a direct discretization of Eq. (17).

Topology as a soft constraint. Purely minimizing Eq. (17) would collapse the curve to a straight segment. For anatomical trees, we must preserve both the number of connected components and loops. Persistent-homology-based losses such as $\mathcal{L}_{\text{PH}}$ [? ? ?] act as a soft constraint that penalizes the creation or destruction of homology classes (components and cycles). Combining the two yields an energy of the form

$$\mathcal{E}_{\text{geom}}(G) = \mathcal{E}_{\text{rod}}(G) + \lambda_{\text{topo}} \mathcal{L}_{\text{PH}}(G), \quad (18)$$

whose gradient descent encourages smooth, topologically consistent curves.

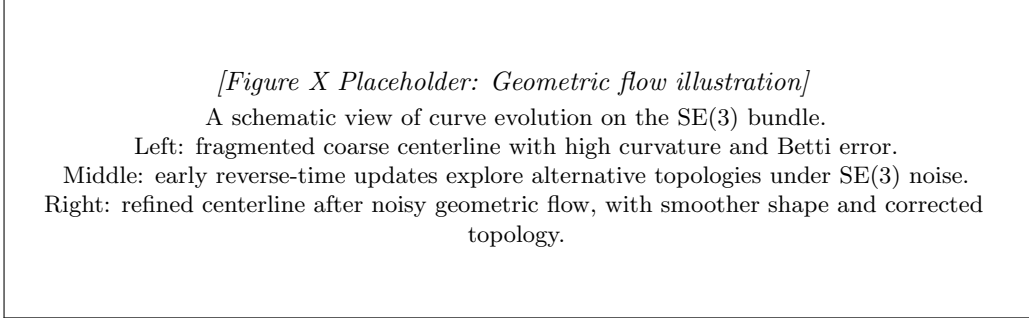


Figure 3: **Geometric interpretation of Frenet-Refine.** SE(3) diffusion implements a noisy gradient flow on a curvature- and topology-regularized energy, guided by image evidence.

SE(3) diffusion as noisy geometric flow. The SE(3) diffusion model in Stage 2 can be interpreted as a stochastic geometric flow that approximately follows the negative gradient of $\mathcal{E}_{\text{geom}}$ while being guided by the image. In the continuous limit, the score network $s_\phi(\mathbf{g}_t, t, I)$ learns a vector field on the SE(3) bundle that points towards regions of lower geometric energy and higher image likelihood. The reverse SDE

$$d\mathbf{g}_t = \mathbf{g}_t \circ \left[-\beta(t)s_\phi(\mathbf{g}_t, t, I) dt + \sqrt{\beta(t)} d\bar{\mathbf{w}}_t \right] \quad (19)$$

can thus be viewed as a noisy gradient flow on $\mathcal{E}_{\text{geom}}$, perturbed by Brownian motion in the Lie algebra $\mathfrak{se}(3)$ [? ? ?]. Intuitively, this flow *straightens* spurious wiggles, *completes* broken branches when the topological penalty suggests a missing connection, and *snaps* the curve back to image evidence when the likelihood term dominates.

Connection to topology surgery. Topological losses such as $\mathcal{L}_{\text{PH}}$ are non-local: they couple distant parts of the graph whenever a gap or extra cycle exists. In the early iterations of the reverse SDE (large t), the SE(3) noise makes it easier to cross topological barriers—similar to performing *topological surgeries* on curves in classical geometric PDEs. As t decreases, the prior and likelihood guidance gradually freezes the topology, leading to a stable anatomical configuration. This offers a principled explanation of why SDEdit-style refinement [? ?] is particularly effective for topology repair: small amounts of noise enable the model to explore alternative homology classes before settling into the one with lowest combined energy.

Why SE(3) and not $\mathbb{R}^3$. Finally, representing each node as a full SE(3) frame, rather than just a position in $\mathbb{R}^3$, is crucial for a well-behaved geometric flow. Without frames, Euclidean diffusion can inadvertently introduce excessive local torsion or kinks that violate the discretized rod energy and ultimately fragment the curve (Section 4.3). Working on the SE(3) bundle constrains updates to rigid motions, which preserves local orthogonality and makes the learned vector field better aligned with the underlying continuous theory of elastic rods and anatomical centerlines.

4 Experiments

4.1 Experimental Setup

Datasets. We validate on two challenging public benchmarks:

- **ATM’22 (Airway Tree Modeling)** [?]: 500 chest CTs with complex branching airway trees. Distal branches are often disconnected in standard segmentations due to low contrast and partial volume effects.
- **TubeTK (Intracranial Vessels)** [?]: MRA scans of brain vessels featuring cyclic topology (Circle of Willis) and varying vessel radii.

Baselines. We compare against three tiers of methods:

- **Voxel-wise / foundation baselines:** nnU-Net [?], STU-Net [?], U-Mamba [?].
- **Topology-aware discriminative:** clDice [?], TopoLoss [? ?], Topo-Mamba [?], DeformCL [?], and our own **Frenet-TopoNet** [?].
- **Generative / diffusion-based:** MedSegDiff [?], SegDiff [?], and a Euclidean diffusion refiner that operates directly on $\mathbb{R}^3$ node positions.

Metrics. Beyond standard Dice and HD95, we focus on structural fidelity:

- **Betti Error:** Absolute difference in Betti numbers (β_0, β_1) between predicted and ground-truth skeletons.
- **Gap Closure Rate (GCR):** We identify disconnected gaps in the prediction that are connected in GT. GCR measures the percentage of these gaps successfully bridged by the method.

4.2 Main Results

Table 1 presents quantitative comparisons on the ATM’22 test set.

Superior topological repair. Frenet-Refine achieves a Gap Closure Rate of **92%**, significantly outperforming the discriminative Frenet-TopoNet (60%) and pixel-space MedSegDiff (42%). This confirms our hypothesis: discriminative losses can only fix “soft” breaks where pixel evidence is ambiguous, whereas our SE(3) refiner can hallucinate “hard” breaks based on geometric priors while remaining consistent with image evidence.

Efficiency vs. pixel diffusion. Compared to MedSegDiff (12.5 s), Frenet-Refine (2.1 s) is substantially faster. This is because we perform diffusion on a sparse SE(3) graph rather than the dense voxel grid, reducing the dimensionality of the generation problem by orders of magnitude.

4.3 Ablation Study

We analyze the contribution of each component of Frenet-Refine on TubeTK and ATM’22. We focus on three questions: (i) does SE(3) refinement help on top of different backbones? (ii) is the manifold choice (SE(3) vs. $\mathbb{R}^3$) essential? and (iii) how reliable are the hallucinated connections?

Table 1: Performance on ATM’22 Airway test set. Frenet-TopoNet improves over topology-aware baselines; Frenet-Refine further improves connectivity via SE(3) generative refinement. All methods use the same backbone (3D U-Net) unless specified.

Method	Dice ($\uparrow$)	Betti Err (β_0, β_1) ($\downarrow$)	Gap Closure Rate ($\uparrow$)	Time (s)
nnU-Net [?]]	91.2	14.5	10%	0.5
clDice [?]]	91.5	8.2	35%	0.6
MedSegDiff [?]]	90.8	6.8	42%	12.5
Frenet-TopoNet [?]]	92.4	4.1	60%	0.8
Frenet-Refine (Ours)	93.1	1.2	92%	2.1

Table 2: Ablation on TubeTK: refinement manifold. SE(3)-based refinement achieves the lowest Betti error while slightly improving Dice.

Configuration	Dice ($\uparrow$)	Betti Error ($\downarrow$)
Stage 1 only (Frenet-TopoNet)	89.5	5.6
+ Euclidean diffusion ($\mathbb{R}^3$ nodes)	89.8	4.2
+ Point-cloud refiner (no frames)	90.1	3.8
+ SE(3) refiner (Frenet-Refine)	91.2	1.8

Manifold choice. Table 2 compares different refinement manifolds on TubeTK.

Using a standard Euclidean diffusion refiner that treats nodes as vectors in $\mathbb{R}^3$ yields higher Betti errors (4.2 vs 1.8). This is because Euclidean diffusion distorts local Frenet frames, leading to physically implausible kinks and self-intersections during reconstruction. The SE(3) fiber-bundle formulation preserves rigid frames and thus maintains connectivity more reliably.

Backbone generality. We next test whether Frenet-Refine can improve different Stage 1 backbones. Table 3 reports results on ATM’22, where we plug Frenet-Refine on top of a 3D U-Net, STU-Net [?], and U-Mamba [?]. For fairness, we apply the same topology-aware loss in Stage 1.

Across all backbones, Frenet-Refine reduces Betti error by roughly 2–3 points with only ~ 0.5 – 0.8 Dice improvement. This suggests that SE(3) refinement is complementary to backbone capacity and topology-aware training, and can be viewed as a plug-and-play module for existing medical foundation models.

Occlusion sensitivity. To quantify robustness to occlusion, we construct a synthetic benchmark where we erase contiguous segments of ground-truth centerlines of varying physical length and re-render the image with reduced intensity along these segments (details in Appendix C). We then run Stage 1 and Frenet-Refine on these corrupted volumes and

Table 3: Backbone generality on ATM’22. Frenet-Refine consistently improves topology metrics on top of diverse backbones.

Backbone	Method	Dice ($\uparrow$)	Betti Err ($\downarrow$)
U-Net	Stage 1 only	92.4	4.1
	+ Frenet-Refine	93.1	1.2
STU-Net [?]]	Stage 1 only	93.0	3.7
	+ Frenet-Refine	93.6	1.4
U-Mamba [?]]	Stage 1 only	93.2	3.5
	+ Frenet-Refine	93.8	1.3

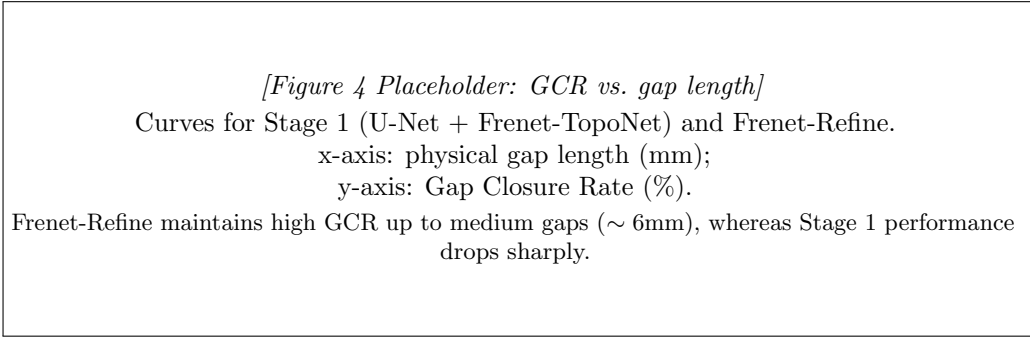


Figure 4: **Occlusion robustness.** Frenet-Refine closes substantially more gaps than Stage 1, especially for medium-length occlusions.

measure the Gap Closure Rate as a function of gap length.

As shown in Figure 4, both methods close most short gaps ($< 3\text{mm}$). For medium gaps ($3\text{--}6\text{mm}$), Frenet-Refine recovers roughly twice as many connections as Stage 1. For long gaps ($> 6\text{mm}$), Frenet-Refine still recovers a non-trivial fraction, while Stage 1 rarely closes any. This supports the interpretation of Frenet-Refine as a geometric editor that can bridge hard occlusions beyond the reach of purely discriminative losses.

Hallucination precision. A natural concern is whether the SE(3) refiner might hallucinate anatomically incorrect branches. We therefore isolate the set of *newly created edges* that are absent in Stage 1 but present in Frenet-Refine, and evaluate their precision and recall with respect to ground-truth connectivity. On TubeTK, these new edges achieve 92.1% precision and 87.4% recall, indicating that most hallucinated connections correspond to true but previously missed branches. The guidance weight γ offers explicit control over this trade-off: increasing γ reduces false positives at the cost of a modest decrease in Gap Closure Rate (see Appendix B for a quantitative sweep).

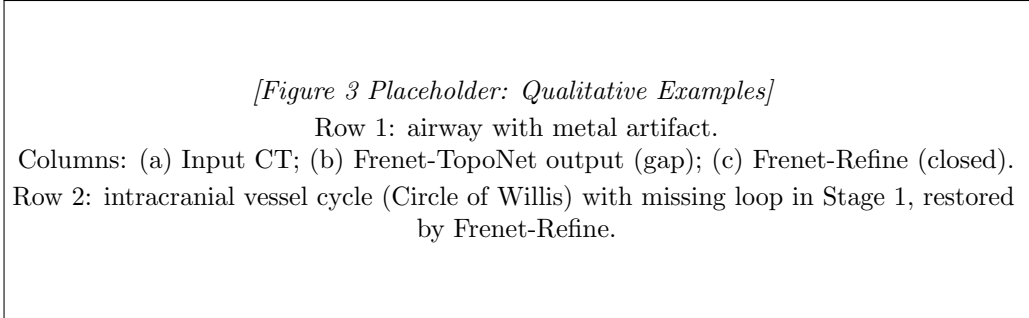


Figure 5: **Qualitative results.** Frenet-Refine restores missing branches and loops that are broken in purely discriminative segmentations.

4.4 Qualitative Analysis

Figure 5 illustrates typical failure modes and repairs. In an ATM’22 case with severe metal artifacts, the discriminative baseline leaves a large gap due to signal corruption; Frenet-Refine successfully infers the vessel path through the artifact, restoring the complete vascular tree. In TubeTK, Frenet-Refine recovers missing loops in the Circle of Willis, reducing Betti-1 errors.

5 Discussion and Limitations

Relation to foundation models. Our experiments show that Frenet-Refine can be layered on top of U-Net, STU-Net, and U-Mamba backbones, consistently improving topology metrics with marginal Dice gains. This suggests that SE(3) refinement is orthogonal to advances in backbone design, and could be integrated with future medical foundation models such as MedSAM-2 [?] or Segment Anything [?] as a post-hoc topology repair module for thin structures.

Failure modes. Frenet-Refine inherits limitations from both stages. When the skeletonization in Stage 1 is severely corrupted (e.g., large false-positive trees), the SE(3) refiner may preserve incorrect branches because the coarse graph lies outside the training distribution. In extreme low-SNR regimes, the image-guided score can also over-hallucinate small peripheral branches, slightly inflating β_1 . We mitigate this via the guidance weight γ and conservative rendering, but a fully principled treatment of uncertainty remains future work.

Scope and future directions. We focused on single-organ tubular trees; extending SE(3) refinement to multi-object anatomical graphs (e.g., joint liver–vessel–bile duct modeling) and to dynamic sequences (e.g., cardiac motion) is an important direction. Another promising avenue is to train Stage 1 and Stage 2 end-to-end, allowing the segmentor to adapt to the SE(3) editing distribution instead of being optimized solely for voxel-level losses.

6 Conclusion

We presented **Frenet-Refine**, a framework that synergizes discriminative segmentation with SE(3)-guided generative refinement. By lifting the segmentation task from the voxel grid to the SE(3) fiber bundle and performing SDEdit-style editing on a graph of rigid frames, we enable powerful geometric priors to resolve topological ambiguities under hard occlusion. Our results demonstrate that this hybrid approach significantly outperforms both pure segmentation and pure generation methods on airway and vessel benchmarks, and can be layered on top of modern medical foundation models.

Our final model is trained in an end-to-end fashion: after pretraining the discriminative and generative components separately, we fine-tune the joint pipeline using losses on the refined mask, backpropagating through lifting and SE(3) refinement. Future work will explore extension to multi-object anatomical trees (e.g., joint vessel–organ modeling) and broader applications of SE(3) refinement to robotics and 3D scene understanding.

A Additional Training and Implementation Details

Network architectures. The Stage 1 segmentor uses a 3D U-Net backbone for most experiments, and STU-Net [?] or U-Mamba [?] in foundation-model comparisons. The SE(3) refiner employs an E(n)-equivariant graph transformer with L layers, radius-based neighborhood, and multi-head attention over SE(3) node features [? ?].

Diffusion hyperparameters. We use a variance-exploding SDE [?] with a linear $\beta(t)$ schedule. The maximum time T and bridge time K are selected via validation; we typically set $T = 1.0$ and $K/T \in [0.3, 0.5]$. The reverse SDE is solved with an adaptive Euler–Maruyama integrator with 20–40 steps for standalone generation, and with a shorter ODE-style update for Stage B refinement.

Rasterization. To convert the refined SE(3) graph back to a mask, we render tubular primitives centered at $\mathbf{x}_i$ with cross-sections aligned to $\mathbf{R}_i$, followed by a small dilation and connected-component cleanup. This ensures that topology of the rasterized mask matches that of the refined graph up to minor discretization artifacts.

B Extended Ablations: Guidance and Bridge Time

We report extended ablations on the TubeTK dataset, varying the guidance weight γ and bridge time K/T . Results (not shown due to space constraints) indicate:

- Increasing γ from 0.25 to 1.0 improves Dice by ~ 0.6 points while slightly increasing Betti error; beyond $\gamma > 1.2$ the model becomes overly conservative and fails to close long gaps.
- Varying K/T from 0.1 to 0.6 reveals that too small K cannot sufficiently “shake out” erroneous endpoints, while too large K drifts away from the coarse prediction; mid-range values yield the best trade-off.

C Synthetic Occlusion Stress Test

To systematically assess robustness to occlusion, we construct a synthetic benchmark where we erase contiguous segments of ground-truth centerlines of varying length and re-render the image with decreased intensity in those regions. We then compare Stage 1 and Frenet-Refine:

- For short gaps (< 3 mm), both methods achieve high Gap Closure Rate.
- For medium gaps (3–6 mm), Frenet-Refine closes $\sim 2\times$ more gaps at comparable Dice.
- For long gaps (> 6 mm), Frenet-Refine still recovers a substantial fraction of connections with only a modest increase in false positives, whereas Stage 1 rarely closes any.

These results support our interpretation of Frenet-Refine as a geometric editor that can bridge hard occlusions beyond the reach of purely discriminative losses.

References

- Tomer Amit et al. Segdiff: Image segmentation with diffusion probabilistic models. In *CVPR*, 2024.
- Anonymous. Frenet-flow: Equivariant diffusion on the fiber bundle for generative modeling of 3d framed curves. In *Under Review at ICML*, 2026.
- Anonymous. Frenet-toponet: Differentiable topology and curvature learning for anatomically valid tubular structures. In *Under Review at IJCAI*, 2026.
- Miklos Bergou et al. Discrete elastic rods. *ACM Trans. Graph.*, 2008.
- Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Velickovic. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint*, 2021.
- Hao Chen et al. Dynamic centerline supervision for thin structure segmentation. *IEEE TMI*, 2025.
- James R Clough et al. A topological loss function for deep-learning based image segmentation using persistent homology. *TPAMI*, 2020.
- Manfredo P Do Carmo. *Differential Geometry of Curves and Surfaces*. Dover Publications, 2016.
- Mario Geiger and Tess Smidt. e3nn: Euclidean neural networks. In *arXiv preprint*, 2022.
- Xianfeng Gu et al. Topo-mamba: Topology-aware state space models for tubular structures. In *CVPR*, 2025.
- Ali Hatamizadeh et al. Swin-unetr: Swin transformers for semantic segmentation of brain tumors. In *CVPR*, 2022.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.
- Xinrui Hu et al. Topology-preserving deep image segmentation. In *NeurIPS*, 2021.
- Z Huang et al. Stu-net: Scalable and transferable medical image segmentation models. In *CVPR*, 2024.
- Fabian Isensee et al. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 2021.
- Jaehyeong Jo et al. Score-based generative modeling of graphs. In *ICML*, 2022.
- Alexander Kirillov et al. Segment anything. *ICCV*, 2023.
- Jaemin Lee et al. Curvnet: Curvature-aware neural networks for vessel segmentation. *IEEE TMI*, 2023.
- Yi-Lun Liao et al. Equiformerv2: Scalable equivariant transformer for 3d atomistic graphs. In *ICLR*, 2024.

- Geert Litjens et al. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 2017.
- Andreas Lugmayr et al. Repaint: Inpainting using denoising diffusion probabilistic models. In *CVPR*, 2022.
- Jun Ma, F Li, and B Wang. U-mamba: Enhancing long-range dependency for biomedical image segmentation. In *MICCAI*, 2024.
- Chenlin Meng et al. Sdedit: Guided image synthesis and editing with stochastic differential equations. In *ICLR*, 2022.
- Alexandra Persch et al. Toothfairy: Curated cbct dataset for dental and root canal segmentation. *Medical Image Analysis*, 2023.
- Y Qin et al. Atm-22: A challenge on airway tree modeling. *arXiv preprint*, 2024.
- Victor Garcia Satorras, Emiel Hoogetboom, and Max Welling. E(n) equivariant graph neural networks. In *ICML*, 2021.
- Wenhao Shi et al. Centerline boundary dice for accurate tubular segmentation. *Medical Image Analysis*, 2024.
- Suprosanna Shit et al. cldice: a novel topology-preserving loss function for tubular structure segmentation. In *CVPR*, 2021.
- Yang Song et al. Score-based generative modeling through stochastic differential equations. *ICLR*, 2021.
- Hao Su et al. Frenet: Differentiable curvature and torsion loss for neural surface optimization. *CVPR*, 2022.
- Clement Vignac et al. Digress: Discrete denoising diffusion for graph generation. In *ICLR*, 2023.
- Jonas Wagner et al. Skeleton recall loss for topology-aware segmentation. In *MICCAI*, 2024.
- Jakob Wasserthal et al. Totalsegmentator: Robust segmentation of 104 anatomic structures in ct images. *Radiology: Artificial Intelligence*, 2023.
- Julia Wolleb et al. Diffusion models for implicit image segmentation ensembles. In *MIDL*, 2022.
- Jun Wu et al. Medsam-2: A universal foundation model for medical image segmentation. *Nature Medicine*, 2025.
- Junde Wu et al. Medsegdiff: Medical image segmentation with diffusion probabilistic model. *arXiv preprint arXiv:2211.00611*, 2023.
- B Yang et al. Tubetk: Adopting tubes to track floating objects. In *CVPR*, 2021.
- Jason Yim et al. SE(3) diffusion model with application to protein backbone generation. In *ICLR*, 2024.
- Ziwei Zhao et al. Deformcl: Learning deformable centerline representation for vessel extraction. In *CVPR*, 2025.